

TABLE 3
INTEL ELKHART LAKE VS INTEL TIGER LAKE

Feature	Intel Elkhart lake	Intel Apollo Lake
CPU specs		
CPU Cores / threads	Up to 4/4	Up to 4/4
Base Frequency	Up to 1.8 GHz	Up to 1.6 GHz
Burst frequency	Up to 3.0 GHz	Up to 2.0 GHz
L2 Cache	1.5 MB	2.0 MB
Silicon Technology	10 nm	14 nm
TDP	Up to 12 W	Up to 12 W
Memory specs		
Max memory	Up to 32 GB	Up to 8 GB
Memory Type	LPDDR4 up to 4267MT/s	LPDDR4 up to 2400 MT/s
ECC –support	IBECC	yes
Graphics		
Graphics	Intel UHD-graphics	Intel HD-Graphics 505
Base frequency	500 MHz	500 MHz
Burts frequency	750 MH	650 MHz
Execution units	32	18
I/Os		
PCIe Gen	PCIe Gen 3	PCIe Gen 2
PCIe Lanes	Up to 8	Up to 6
SATA	2x	2x
USB support	2.0 / 3.0 / 3.1	2.0 / 3.0
USB ports	4	8
Integrated Ethernet	3x 2,5 GbE	-

is very important for the industrial segment.

And with the introduction of Elkhart Lake, embedded applications can now take advantage of Intel's 10nm SuperFin technology, which made its debut with this low-power Atom processor technology. Increased RAM support offers far more memory bandwidth and allows for much higher transfer rates. Up to 32GB LPDDR4 memory with data rates reaching 4267 MT/s, in contrast to 8GB LPDDR4 with up to 2400 MT/s from the previous generation, are now supported. The possibility to add additional memory capabilities with Intel's budget-friendly Inband Error Correction Code (IBECC) is also a welcomed advantage for mission-critical real-time applications. Instead of using expensive, dedicated ECC memory modules, designers can leverage

lower-cost conventional RAM. Setting the mode as ECC or non-ECC can be easily carried out in the BIOS, which means developers can choose which areas of memory will be assigned as IBECC.

Highly accelerated graphics performance with a small appetite

The latest generation of processors makes use of Intel's Gen11 graphics units, located on top of the CPU die, which double graphics performance to a maximum speed of 3x 4k @ 60fps and 10-bit colour depth. Once again, the use of 10nm technology in the manufacturing process allows for stunning high performance with low power requirements. The greatest contributor to the doubling of performance is the increase of up to 32 integrated execution units (EUs) in the Gen11 graphics unit, com-

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